

(55) Вычислить  $\int_0^{+\infty} x^{2k} e^{-ax^2} dx$ .

Решение леммы.

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$$2^{2\alpha+1} \Gamma(\alpha) \Gamma(\alpha + \frac{1}{2}) = \sqrt{\pi} \Gamma(2\alpha)$$

Пример 1

$$\int_0^{+\infty} x^{2k} e^{-ax^2} dx = \left| \begin{array}{l} x\sqrt{a} = \sqrt{t} \Rightarrow x = \sqrt{\frac{t}{a}} \\ dx = \frac{1}{2}(\frac{t}{a})^{-\frac{1}{2}} \frac{1}{a} dt = \frac{1}{\sqrt{a}} \frac{dt}{2\sqrt{t}} \\ -ax^2 = -t \end{array} \right|$$

|     |     |           |
|-----|-----|-----------|
| $x$ | $0$ | $+\infty$ |
| $t$ | $0$ | $+\infty$ |

$$= \frac{1}{(\sqrt{a})^{2k}} \int_0^{+\infty} t^k e^{-\frac{t}{2\sqrt{a}}} \frac{dt}{2\sqrt{t}} = \frac{1}{2(\sqrt{a})^{2k+1}} \int_0^{+\infty} t^{k-\frac{1}{2}} e^{-\frac{t}{2\sqrt{a}}} dt$$

$$= \frac{1}{\sqrt{a}^{2k+1}} \cdot \frac{1}{2} \Gamma(k + \frac{1}{2}) \quad (\equiv)$$

Вспомог.  $\Gamma(k + \frac{1}{2})$

$$\Gamma(\alpha+1) = \alpha \Gamma(\alpha), \quad \Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$\Gamma(k + \frac{1}{2}) = k + \frac{1}{2} \Gamma(k - \frac{1}{2}) = \dots = \frac{(2k-1)!!}{2^k} \Gamma(\frac{1}{2})$$

$$= \frac{(2k-1)!!}{2^k} \sqrt{x}$$

$$\Leftrightarrow \frac{\sqrt{x}}{\sqrt{a}^{2k+1}} \frac{(2k-1)!!}{2^{k+1}} \leftarrow \text{Ombet}$$

